

How Do Users Review Spotify?

Analyzing Emotion, Topics, and Platform Replies Using Text as Data and Logistic Regression

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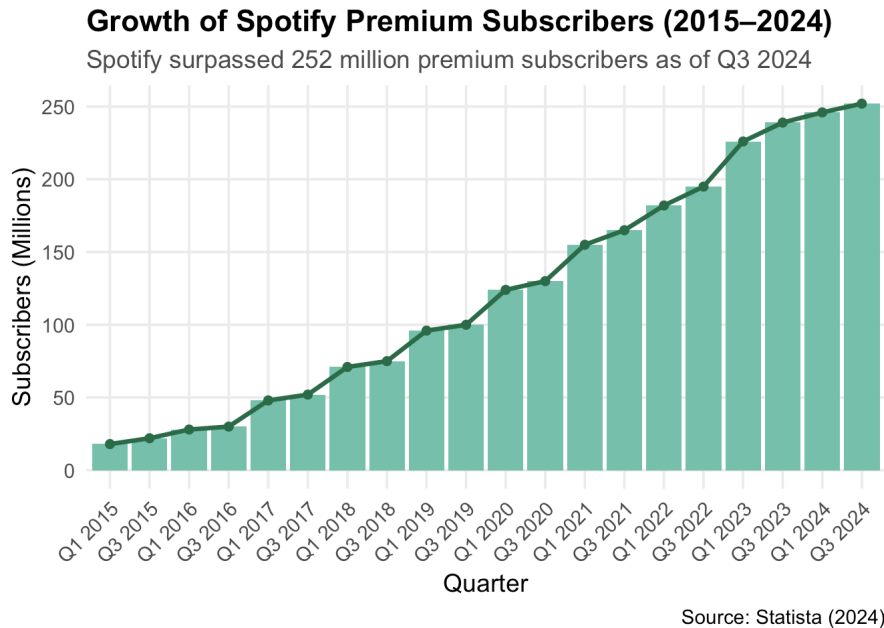
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1. Introduction

Spotify has been the most popular music streaming platform. On the Google Play Store, users leave public reviews about the app—yet only a few ever receive an official reply from Spotify.



Goals:

- **(1) Business Strategy Perspective:** Analyze user-generated reviews to get key experience themes and sentiment patterns, to improve app functionality, user satisfaction, and retention.
- **(2) Digital Platform Governance Perspective:** Provide insights into Spotify's public engagement practices, including reply prioritization and transparency considerations.

1.1 Research Questions

- **RQ1:** How do Spotify users describe their **experience** in written reviews?

H1: Most reviews focus on **app functionality, usability, and technical performance** rather than music content.

- **RQ2:** Which review **topics or tones** are more likely to receive an **official reply**—and what does this reveal about platform priorities or transparency?

H2: Reviews that express **dissatisfaction**, particularly those mentioning **technical issues or account problems**, are more likely.

1.2 Dataset

- 61,594 Spotify user reviews scraped from Google Play (2022 Kaggle dataset).
- Key fields collected:
 - **Star Rating** (1–5)
 - **Review Text**
 - **Review Timestamp**
 - **Number of Thumbs-up**
 - **Official Reply** (if any)

Deeper information needed for meaningful analysis is not directly available. Specifically:

- **Review Categories:** Does the review focus on App Experience or something else (e.g., music content)?
- **Review Topics:** Is the review about technical issues, UX features, account problems, or other concerns?
- **Sentiment and Emotions:** Does the review convey joy, anger, frustration, or satisfaction?
- **Reply Status:** Which reviews received an official response (0/1)?

1.3 Methodology

- **Manual Annotation:**

Hand-labeled 2,000 reviews into categories (e.g., App Experience vs. Other) for supervised learning.

- **Supervised Machine Learning:**

- Models: Naïve Bayes, SVM (linear kernel), and Lasso (logistic regression)
- Features: TF-IDF weighted matrices and GloVe word embeddings
- Cross-validation: 5-fold CV for hyperparameter tuning (e.g., SVM cost, Lasso lambda)

- **Unsupervised Topic Modeling:**

- LDA (Latent Dirichlet Allocation) applied to App Experience reviews
- Number of topics selected based on interpretability ($K = 6$)

- **Sentiment Analysis:**

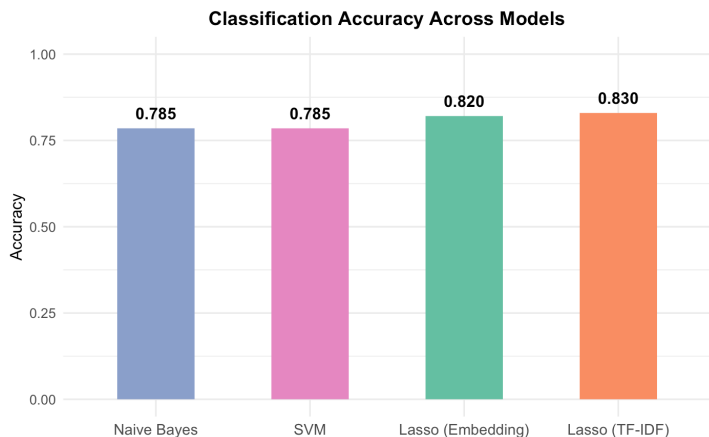
- NRC lexicon mapping to classify emotions (e.g., anger, joy, sadness) at the review level

- **Logistic Regression Analysis:**

- Investigated predictors of official reply likelihood using review sentiment, topic, and length

2. Labeling Results

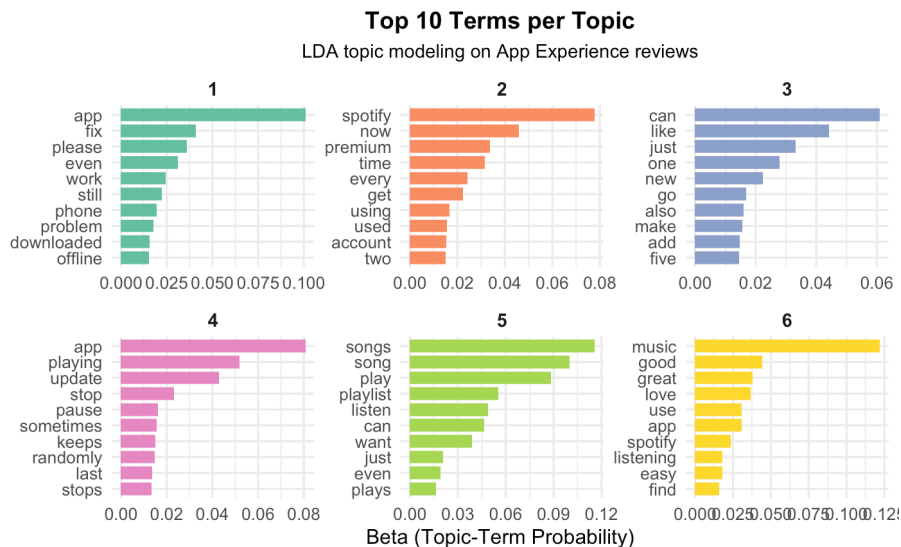
2.1 Supervised Machine Learning: Review Category Prediction



Metric	TF-IDF Lasso	Word Embedding Lasso
Accuracy	0.83	0.82
Kappa	0.648	0.613
Precision	0.868	0.837
Recall	0.846	0.878
F1 Score	0.857	0.857
Balanced Accuracy	0.826	0.802

- **TF-IDF Lasso** slightly outperformed Word Embedding Lasso across multiple evaluation metrics.
- Selected **TF-IDF Lasso** for full dataset classification based on higher accuracy and better balanced performance.

2.2 Unsupervised Topic Modeling: Sub-Themes in App Experience Reviews



Topic 1 : Technical Fixes



Topic 2 : Account & Subscription



Topic 3 : General UX Feedback



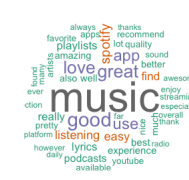
Topic 4 : Playback Issues



Topic 5 : Music and Playlist Feature



Topic 6 : Positive Experience



- **LDA Topic Modeling (K = 6)** revealed key sub-themes within App Experience feedback.

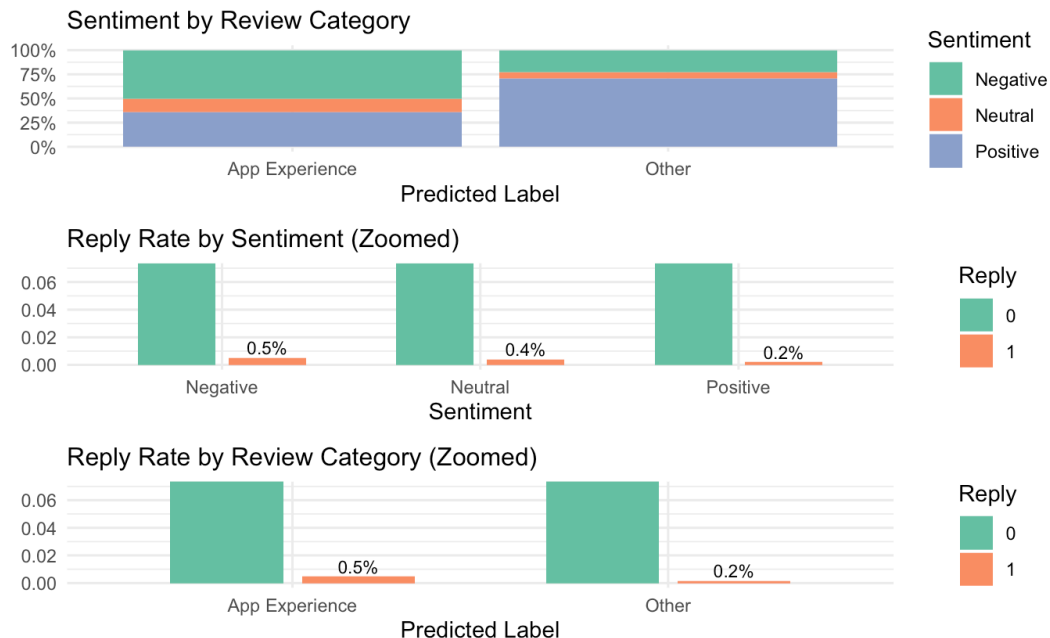
3. Descriptive Overview

3.1 Sentiment, Reply Rate, Label



- Most reviews are **positive** in tone.
- Only **3.5%** of reviews receive an official reply.
- Around **60%** of all reviews are classified as **App Experience**.

3.2 Sentiment-Reply Patterns by Category

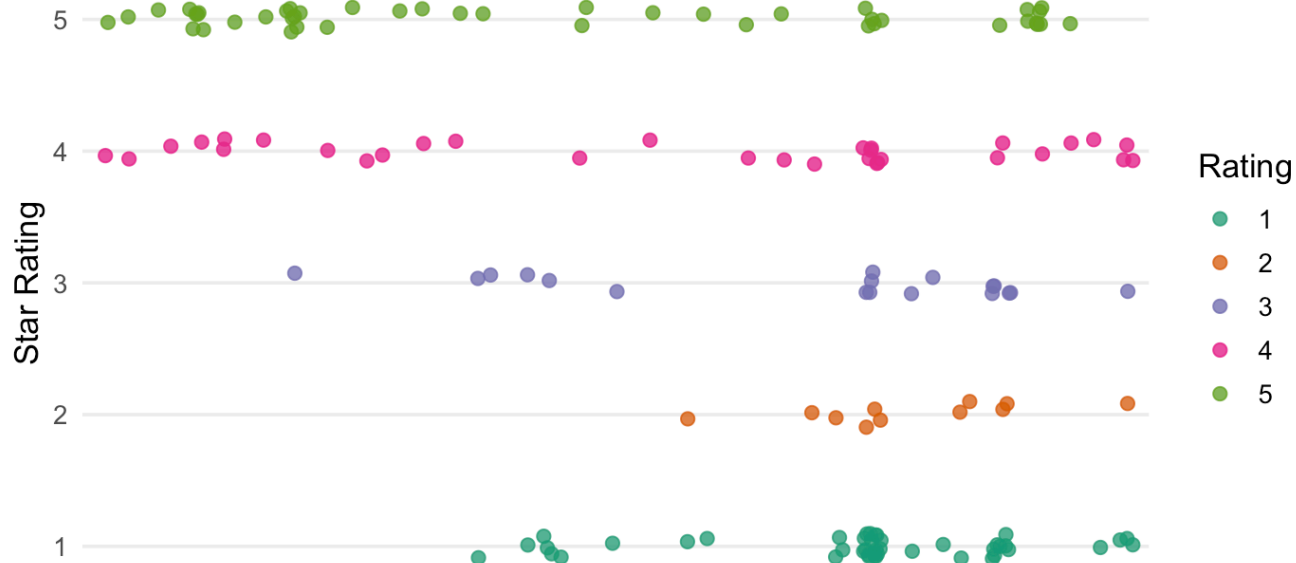


- **App Experience** reviews are more likely to have negative star ratings.
- **Negative** reviews are more likely to receive a reply than **positive** ones.
- Reviews about **App Experience** are more likely to be replied to than reviews about other topics.

3.3 Duplicate Reviews

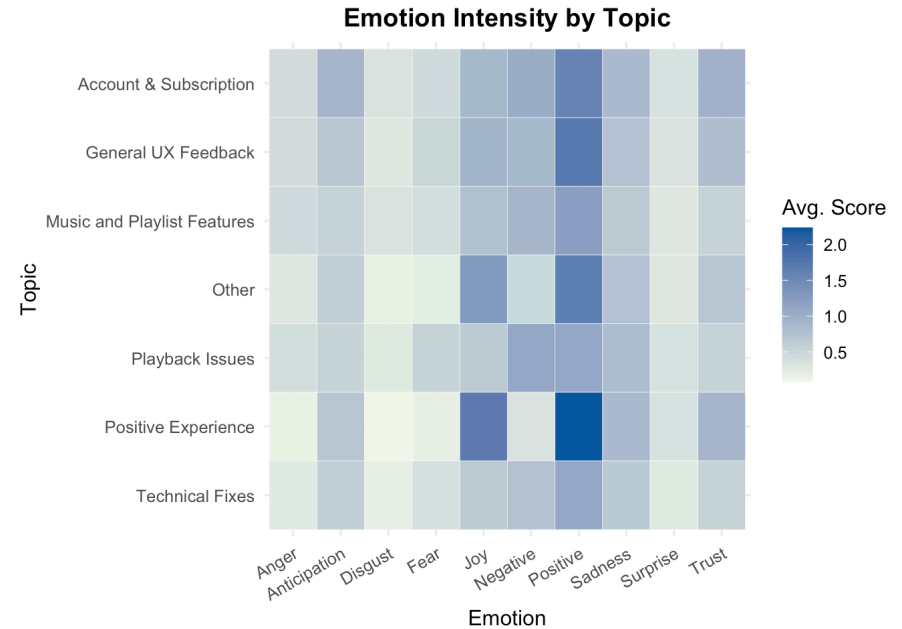
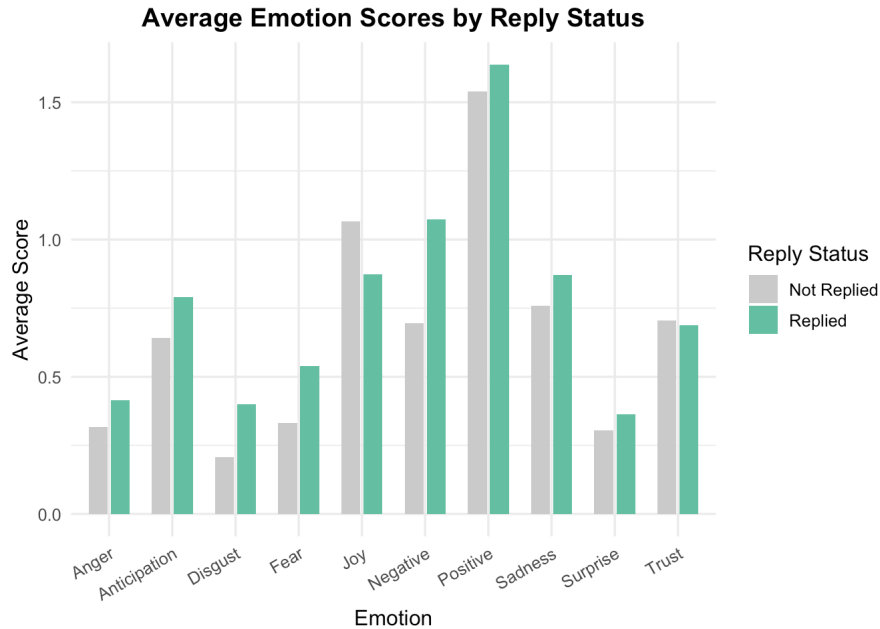
Variation in Star Ratings for Identical Reviews

Identical reviews submitted with different star ratings



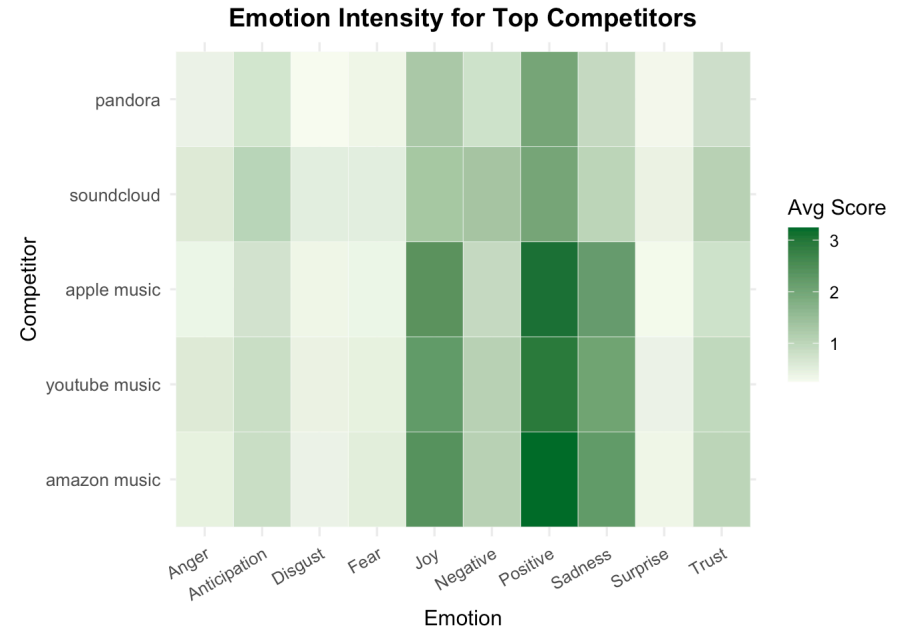
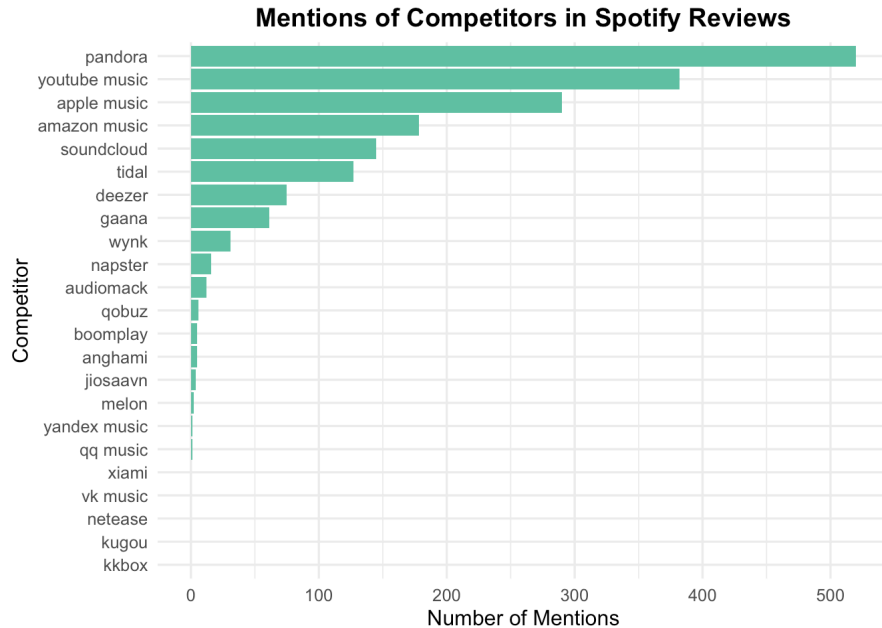
- One interesting observation is that the same review can get different ratings.

3.4 Emotion Score



- **Disgust, Fear, and Negative** emotions are **higher** in reviews that received a **reply**.
- Reviews expressing **Joy** and **Trust** are **less likely** to receive a reply.

3.5 Competitors



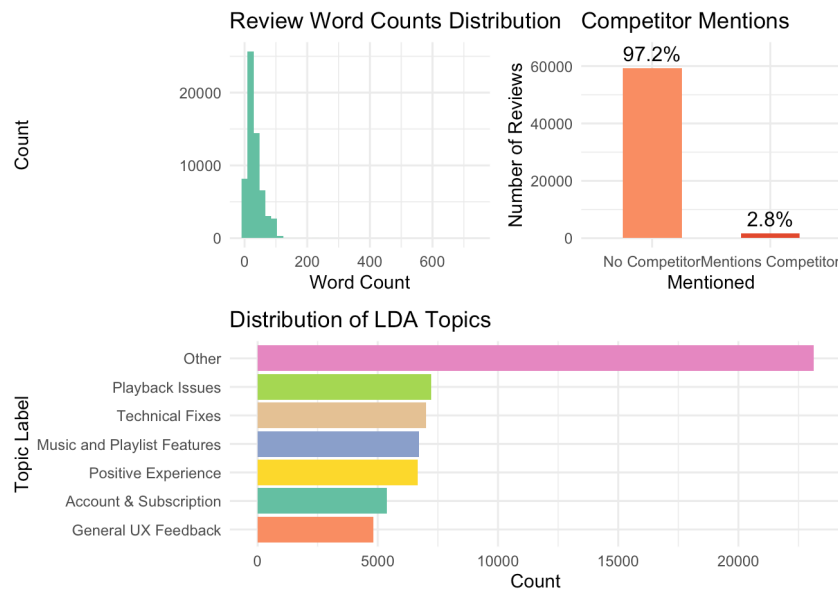
- Apple Music, YouTube Music and Amazon Music clearly dominate in emotional engagement (**joy**, **positive** and **sadness**).

4. Reply Analysis

4.1 Logistic Regression

$$\begin{aligned} \text{logit}(\Pr(\text{Replied}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{anticipation}_i + \beta_2 \cdot \text{anger}_i + \beta_3 \cdot \text{disgust}_i + \beta_4 \cdot \text{fear}_i \\ & + \beta_5 \cdot \text{joy}_i + \beta_6 \cdot \text{sadness}_i + \beta_7 \cdot \text{surprise}_i + \beta_8 \cdot \text{trust}_i \\ & + \beta_9 \cdot \text{negative}_i + \beta_{10} \cdot \text{positive}_i + \beta_{11} \cdot \text{word_count}_i \\ & + \beta_{12} \cdot \text{competitor_mention}_i + \beta_{13} \cdot \text{PredictedLabel_LDA}_i + \varepsilon_i \end{aligned}$$

Are emotion, word count, competitor-mention, and topic labels linked to whether a user gets a Spotify reply?



4.2 Regression Result

Table 1: Logistic Regression: Predicting If a Review Receives a Spotify Reply

Variable	Category	Coefficient	Significance
(Intercept)	—	-5.528	***
anticipation	NRC Lexicon	0.049	
anger	NRC Lexicon	-0.283	*
disgust	NRC Lexicon	0.464	***
fear	NRC Lexicon	0.086	
joy	NRC Lexicon	-0.240	*
sadness	NRC Lexicon	-0.079	
surprise	NRC Lexicon	0.107	
trust	NRC Lexicon	-0.226	*
negative	NRC Lexicon	-0.017	
positive	NRC Lexicon	0.179	*
word_count	Other Controls	0.010	**
competitor_mention	Other Controls	-0.103	
General UX Feedback	Topic Label	0.076	
Music and Playlist Features	Topic Label	-0.717	*
Other	Topic Label	-1.031	***
Playback Issues	Topic Label	0.176	
Positive Experience	Topic Label	-0.915	**
Technical Fixes	Topic Label	-0.288	
Observations	60,915		
Log Likelihood	-1365.859		
AIC	2769.7		
McFadden R^2	0.044		

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

NRC Lexicon variables are derived from the NRC Emotion Lexicon.

Topic Label reference category is **Account & Subscription**.

Coefficients represent changes in log-odds of receiving a Spotify reply.

- Spotify is more likely to reply to **longer reviews** and those expressing **disgust**.
- **Joyful** or **trusting** tones reduce reply likelihood.
- Reviews about **technical** and **account** issues get more attention.
- Reviews on **music features** or general **praise** are often ignored.

4.3 SMOTE Adjustment for Reply Imbalance

To evaluate whether findings are robust to imbalance, I re-estimated the reply model using SMOTE-balanced data.

Table 2: Logistic Regression: Predicting Spotify Replies (Original vs. SMOTE Adjusted)

Variable	Category	Original Model	SMOTE Adjusted Model
(Intercept)	—	-5.528*** (0.207)	0.011 (0.021)
anticipation	NRC Lexicon	0.049 (0.090)	0.142*** (0.011)
anger	NRC Lexicon	-0.283* (0.137)	-0.202*** (0.016)
disgust	NRC Lexicon	0.464*** (0.138)	0.371*** (0.017)
fear	NRC Lexicon	0.086 (0.119)	-0.106*** (0.014)
joy	NRC Lexicon	-0.240* (0.121)	-0.609*** (0.014)
sadness	NRC Lexicon	-0.079 (0.107)	0.069*** (0.012)
surprise	NRC Lexicon	0.107 (0.130)	0.070*** (0.014)
trust	NRC Lexicon	-0.226* (0.105)	-0.212*** (0.011)
negative	NRC Lexicon	-0.017 (0.096)	-0.062*** (0.011)
positive	NRC Lexicon	0.179* (0.074)	0.160*** (0.009)
word_count	Other Controls	0.010** (0.003)	0.023*** (0.000)
competitor_mention	Other Controls	-0.103 (0.426)	-0.746*** (0.046)
General UX Feedback	Topic Label	0.076 (0.252)	0.019 (0.025)
Music and Playlist Features	Topic Label	-0.717* (0.288)	-0.395*** (0.025)
Other	Topic Label	-1.031*** (0.250)	-0.934*** (0.022)
Playback Issues	Topic Label	0.176 (0.229)	-0.148*** (0.024)
Positive Experience	Topic Label	-0.915** (0.349)	-0.560*** (0.027)
Technical Fixes	Topic Label	-0.288 (0.255)	-1.098*** (0.027)
Observations		60,915	121,330
Log Likelihood		-1365.859	-74247.462
RMSE		0.06	0.46
McFadden R^2		0.044	0.117

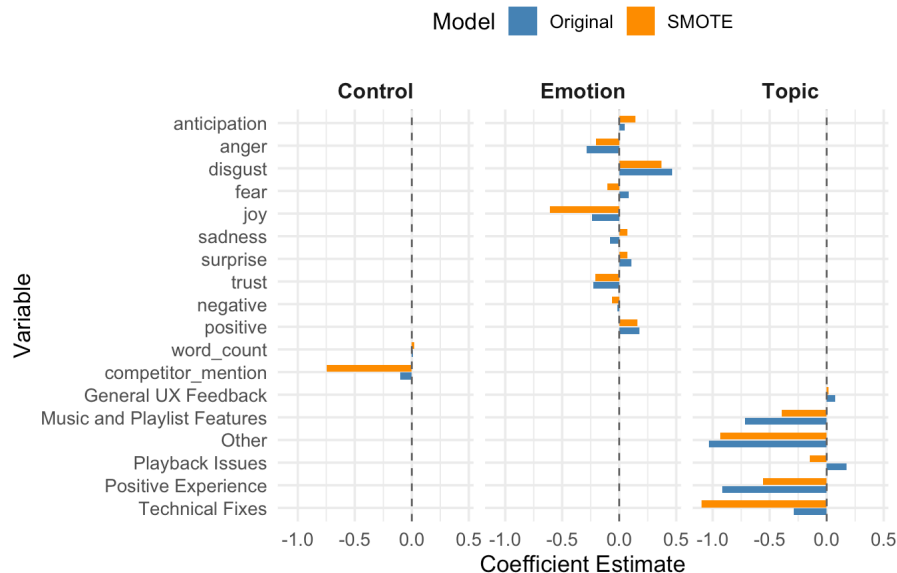
Notes: Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

NRC Lexicon variables are derived from the NRC Emotion Lexicon.

Topic Label reference category is **Account & Subscription**; updated for both models.

Coefficients represent changes in log-odds of receiving a Spotify reply.

Comparison of Coefficient Estimates



5. Discussion

-  Most users complain about **app function**, not music (**RQ1 supported**)
-  Spotify replies mostly to **negative, detailed reviews** (disgust, account issues)
-  **Joyful/trusting reviews** are often ignored—missed opportunity to engage satisfied users
-  Reply strategy is **reactive (problem)**, not proactive (praise) (**RQ2 supported**)
-  SMOTE results confirm key findings are **robust**
-  **Competitor mentions** evoke strong emotion—signal of potential **churn risk**

5.1 Limitations

- **Classification:**

Word embedding models underperformed compared to TF-IDF Lasso; further tuning or larger embedding datasets may improve results.

- **Topic Modeling:**

Some topics show overlap and limited distinctiveness; better preprocessing, validation, or alternative models could enhance separation.

- **Duplicates:**

Although duplicate reviews were retained, their short text length may reduce analysis quality; impact is limited given the small number (~270).

- **Word Clouds:**

Repetitive terms like “song” and “songs” appeared separately - minor stemming or tokenization issues.

- **Regression Sample Size:**

The number of replied reviews was extremely small (<3.5%), limiting statistical power before adjustment.

- **Policy Implications:**

Although reply patterns hint at engagement priorities, deeper exploration is needed to fully assess platform transparency and accountability.

5.2 Lessons Learned and Next Steps

- Understanding models, focusing on details, and building a structured workflow greatly improved project quality
- Future work could explore more advanced models and deeper policy implications

Thank You!